



Extraction of aerodynamic damping and prediction of vortex-induced vibration of bridge deck using CFD simulation of forced vibration

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ABSTRACT

This study presents computational fluid dynamics (CFD) simulation of forced vibration of a bridge deck for extraction of amplitude-dependent aerodynamic damping in the vicinity of vortex lock-in region. The coordinate system fixed on the oscillating body is used to bypass the requirement of mesh updating. The three-dimensional flow around the vibrating deck is assumed to be homogeneous in spanwise direction and simulated using the efficient high-order spectral element/Fourier method. The aerodynamic damping is extracted from the simulated lift and then used to estimate the steady-state vortex-induced vibration (VIV) amplitude for given wind velocity and structural parameters. A good match of VIV amplitudes between simulations and wind tunnel tests validates the accuracy of simulation. The VIV mechanisms are discussed from the vortex shedding modes, distributions of dynamic pressure and energy transfer from flow to bridge deck. The leading-edge vortex shedding rather than Karman vortex shedding is identified to be responsible for the VIV. The hysteresis of VIV is observed which is related to the mode change of impinging leading-edge vortex and leading-edge vortex shedding. Finally, a novel forced vibration technique with a varying amplitude is proposed that permits extraction of aerodynamic damping by a single CFD run with improved efficiency.

1. Introduction

The separating boundary layer flow around a bluff body generally leads to vortex shedding, which imposes oscillating crosswise forces on the body. The phenomenon of lock-in, or synchronization, occurs when the force frequency is close to the body vibration frequency, resulting in possible vortex-induced vibration (VIV). Bridge decks typically have fixed flow separation points and long afterbody. In addition to Karman vortex, the leading-edge vortex shedding can be responsible for the VIV (Shiraishi and Matsumoto, 1983; Nakamura and Nakashima, 1986; Matsumoto et al., 1993; Naudascher and Wang, 1993; Deniz and Staubli, 1997; Cao et al., 2020).

The free and forced vibration wind tunnel tests are used for VIV study of a bridge deck. In the free vibration test, the bridge section is spring-supported, and body motion induced by its interaction with the flow is recorded. In the forced vibration test, the body is controlled to have a prescribed motion, typically a harmonic motion with a constant amplitude, and the resulting forces exerted by the flow are measured. Both free and forced vibration approaches can be used to extract amplitude-dependent aerodynamic damping, and then to predict the

response of structures with various dynamic parameters (Hao et al., 2020; Wang et al., 2020; Wu and Chen, 2020). It was shown that the forced vibration tests can be used to predict accurately the response of freely vibrating bodies (e.g., Morse and Williamson, 2009). It is worth noting that the amplitude-dependence of aerodynamic derivatives of some bridge decks is also evident at wind speed region away from VIV region (e.g., Chen et al., 2005; Wang et al., 2020), which is important especially for predicting the amplitude of soft-type flutter.

The free and forced vibrations are also simulated by computational fluid dynamics (CFD) methods. The free vibration simulation requires the coupling of solvers for flow and structural dynamics to obtain the VIV amplitude (Sun et al., 2009; Álvarez et al., 2019), which generally requires a long duration of simulation as vibration amplitude generally slowly grows with time to steady-state level due to a low level of aerodynamic damping. On the other hand, the forced vibration simulation only requires the flow solver and is relatively easy to be implemented (Sarwar and Ishihara, 2010; Noguchi et al., 2020).

Tamura (1999) reviewed previous numerical studies of aeroelastic wind-structure interaction problems, in which VIV of the rectangular cylinder was predicted from free-vibration simulations with coordinate

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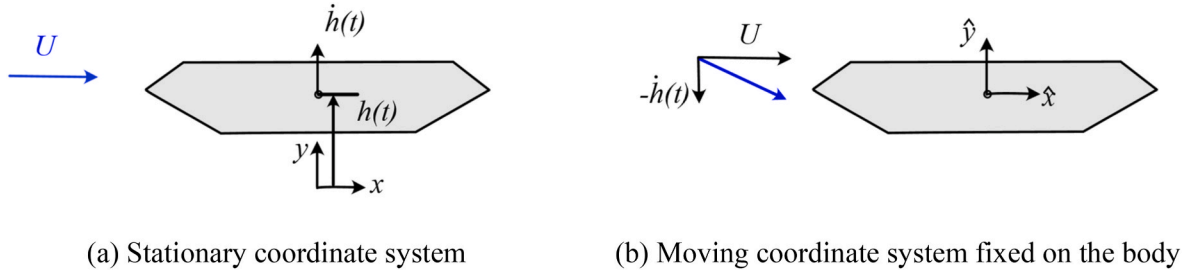


Fig. 1. Forced vibration of the bridge deck.

transformation formulation and finite difference method, where the third-order upwind scheme was applied without employing turbulence models. The effects of initial displacement on VIV response were also discussed in Tamura (1999), which clearly showed the hysteresis characteristic of VIV. Sun et al. (2009) examined the suitability of different turbulence models for simulating aeroelastic problems in the Reynolds Averaged Navier-Stokes (RANS) framework and concluded that $k-\omega$ model is an appropriate choice balancing the computational efficiency and accuracy. The more practical numerical study of bridge deck VIV, e.g., numerical prediction of steady-state VIV amplitude, is more limited, partly due to large computational cost for both free-vibration simulation (long time required to achieve steady-state VIV response) and forced-vibration simulation (large parameter space of reduced velocity and amplitude). Noguchi et al. (2020) predicted the steady-state VIV amplitude of a box girder bridge deck using forced vibration simulation with coordinate transformation in large-eddy simulation (LES). However, the predicted vibration amplitude was much larger than that from free vibration wind tunnel test.

This study uses high-order spectral element/Fourier method with coordinate transformation for forced vibration simulations. The high-order accurate and low-diffusion spectral element method (Patera, 1984; Deville et al., 2002; Karniadakis and Sherwin, 2013) has the potential to accurately compute the vortex-dominated flows (Ekaterinaris, 2005). It is also more efficient than low-order ones, such as finite volume method and low order finite element methods, because a much coarser mesh can be used to achieve same error threshold (Wang et al., 2013). By assuming that the flow is homogeneous in the spanwise direction, the Fourier expansion can be used to represent the three-dimensional (3D) effects with improved computational efficiency (Newman and Karniadakis, 1997). It was also found by Tutar and Holdó (2000) that 3D simulation of a vibrating circular cylinder with a limited spanwise length provides better agreement with sectional experimental data than 2D simulation. The coordinate transformation (Newman and Karniadakis, 1997; Serson et al., 2016) is used to formulate the governing equations in the coordinate system attached to the oscillating body, thus the time-dependent and deforming computational domain is mapped to stationary and non-deforming one. The advantage of such transformation is that a high-quality static mesh can be used without the requirement of mesh updating algorithms essential in, for example, arbitrary Lagrangian-Eulerian (ALE) method (Donea et al., 2017). The spectral element/Fourier method with coordinate transformation is widely used to simulate the VIV of circular cylinders with reasonably good accuracy to capture interested physical quantities, such as the drag and lift coefficients, lift coefficients in phase with velocity (e.g., Newman and Karniadakis, 1997; Dong and Karniadakis, 2005). Current study presents one of the first attempts to apply the spectral element/Fourier method for bridge deck VIV simulations.

The turbulence modeling based on Reynolds-averaged Navier-Stokes equations (RANS) or large eddy simulation (LES) is not capable of calculating the full flow past a stationary cylinder, much less an oscillating one (Al-Jamal and Dalton, 2004). It is also unclear about the effects of numerical dissipation introduced by turbulence modeling to the prediction of aerodynamic damping, which is characterized by small

value and critical for the prediction of VIV amplitude. Based on these considerations, a moderate high Reynolds number close to the target wind tunnel test is chosen in current study to bypass the necessity of turbulence modeling and to make accessible the computational cost.

In this study, the aerodynamic damping of a streamlined bridge deck section, as a function of reduced wind velocity and vibration amplitude, is extracted from simulations of lift force on the deck section under forced vibration. The steady-state VIV amplitude for given wind velocity and structural parameters is then estimated under the condition of zero total system damping (including structural damping and aerodynamic damping), which shows promising prediction accuracy compared with free-vibration wind tunnel test. The VIV mechanisms related to effects of bridge leading barriers and hysteresis phenomenon are discussed from the vortex shedding modes, distributions of dynamic pressure and induced energy transfer on bridge deck surface. Finally, a novel forced vibration technique with a varying amplitude is proposed that permits extraction of aerodynamic damping for a range of vibration amplitude by a single CFD run with greatly improved efficiency.

2. Methodology

2.1. Governing equations

The interaction of incompressible flow with smooth inflow and a vibrating bridge deck is considered as shown in Fig. 1(a). The bridge deck is regarded as a rigid body, and its motion is constrained to move only in the crosswise direction, i.e., y direction, which is denoted as $h(t)$ and t is time variable. The incompressible flow is described through the Navier-Stokes equations. In a stationary and inertial Cartesian coordinate system $(x, y, z; t)$, these equations are given by

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} \quad (1a)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (1b)$$

where ρ is the air density; ν is the fluid kinematic viscosity; $\mathbf{u} = (u, v, w)$ is velocity vector field in terms of flow velocities in streamwise x , crosswise y and spanwise z directions; p is pressure field; $\nabla \equiv (\partial/\partial x, \partial/\partial y, \partial/\partial z)$; and $\nabla^2 \equiv \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$.

To simplify the solution to the flow equations, the moving coordinate system fixed on the body, $(\hat{x}, \hat{y}, \hat{z}; \hat{t})$, is used to transform the time-dependent and deforming flow domain to a stationary and non-deforming one (Newman and Karniadakis, 1997), as seen in Fig. 1(b). The coordinates and velocities in the inertial and noninertial frames are related as

$$\hat{x} = x, \quad \hat{y} = y - h(t), \quad \hat{z} = z, \quad \hat{t} = t \quad (2a)$$

$$\hat{u} = u, \quad \hat{v} = v - \dot{h}(t), \quad \hat{w} = w, \quad \hat{p} = p \quad (2b)$$

According to this transformation, for a function $f(\hat{x}, \hat{y}, \hat{z}, \hat{t})$, we have, for example, $\frac{\partial f}{\partial \hat{t}} = \frac{\partial f}{\partial t} - \dot{h} \frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial \hat{x}} = \frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial \hat{y}} = \frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial \hat{z}} = \frac{\partial f}{\partial z}$. The Navier-Stokes equations in the noninertial frame can be derived as

$$\frac{\partial \hat{\mathbf{u}}}{\partial t} + (\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}} = -\frac{1}{\rho} \hat{\nabla} \hat{p} + \nu \hat{\nabla}^2 \hat{\mathbf{u}} + \hat{\mathbf{F}} \quad (3a)$$

$$\hat{\nabla} \cdot \hat{\mathbf{u}} = 0 \quad (3b)$$

where $\hat{\nabla} \equiv (\partial/\partial \hat{x}, \partial/\partial \hat{y}, \partial/\partial \hat{z})$; and $\hat{\mathbf{F}} = (0, -d\hat{h}/d\hat{t}, 0)$ is the additional body forcing term due to the unsteady coordinate acceleration.

For the boundary conditions, the inlet was imposed with Dirichlet condition on the velocity, and high-order pressure boundary condition (Karniadakis et al., 1991; Serson et al., 2016) on the pressure. The outflow was imposed with Neumann condition on the velocity and Dirichlet condition on the pressure. The wall (bridge deck surface) was imposed with Dirichlet condition (no-slip condition) on the velocity relative to the moving boundary. All Dirichlet boundary conditions specified in the inertial Cartesian (physical) space can be transformed to the noninertial computational frame of reference based on the coordinate mapping.

2.2. Spectral element/Fourier method

To simplify the simulation of the incompressible flow around the vibrating bridge deck, it is assumed that the flow and bridge deck motion are homogeneous in the spanwise direction (z-direction) with a periodic length L_z . Thus, the flow variables in the body-fitted coordinated system can be expanded as

$$\hat{\mathbf{u}}(\hat{x}, \hat{y}, \hat{z}, \hat{t}) = \sum_{m=0}^{M-1} \hat{\mathbf{u}}_m(\hat{x}, \hat{y}, \hat{t}) e^{i\beta m \hat{z}} \quad (4a)$$

$$\hat{p}(\hat{x}, \hat{y}, \hat{z}, \hat{t}) = \sum_{m=0}^{M-1} \hat{p}_m(\hat{x}, \hat{y}, \hat{t}) e^{i\beta m \hat{z}} \quad (4b)$$

where $\beta = 2\pi/L_z$ is the spanwise wavenumber; and M is the number of Fourier modes used in the expansion; $i = \sqrt{-1}$.

The Navier-Stokes equations in the body-fitted noninertial coordinate system become a system of two-dimensional partial differential equations coupled by the nonlinear terms (Newman and Karniadakis, 1997)

$$\frac{\partial \hat{\mathbf{u}}_m}{\partial t} + [(\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}}]_m = -\frac{1}{\rho} \hat{\nabla} \hat{p}_m + \nu (\hat{\nabla}_{xy}^2 - \beta^2 m^2) \hat{\mathbf{u}}_m + \hat{\mathbf{F}} \delta_{m0} \quad (5a)$$

$$\hat{\nabla} \cdot \hat{\mathbf{u}}_m = 0 \quad (5b)$$

where $[(\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}}]_m$ is the coefficient of m th Fourier mode for the nonlinear term, i.e., $(\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}} = \sum_{m=0}^M [(\hat{\mathbf{u}} \cdot \hat{\nabla}) \hat{\mathbf{u}}]_m \exp(i\beta m \hat{z})$; $\hat{\nabla} \equiv (\partial/\partial \hat{x}, \partial/\partial \hat{y}, i\beta m)$, $\hat{\nabla}_{xy}^2 = \partial^2/\partial \hat{x}^2 + \partial^2/\partial \hat{y}^2$; $\delta_{m0} = 1$ when $m = 0$ and $\delta_{m0} = 0$ when $m \neq 0$.

The Navier-Stokes equations are discretized in the $(\hat{x}, \hat{y})$ -plane through a Galerkin spectral element method (Karniadakis and Sherwin, 2013). The 2D spatial domain in the $(\hat{x}, \hat{y})$ -plane is discretized into finite elements, and within each element, high-order Lagrange polynomials based on Gauss-Lobatto-Legendre (GLL) quadrature points are employed

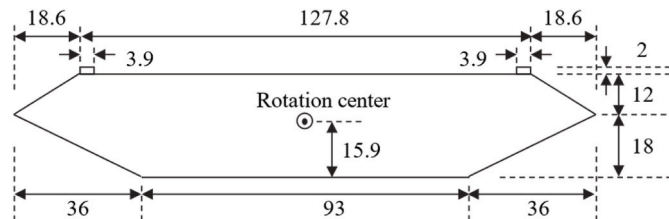


Fig. 2. Bridge deck cross-section (unit: mm) used in wind tunnel test (Noguchi et al., 2020) and current simulation.

as the trial (expansion) and test functions. The Navier-Stokes equations can be discretized in time using the high-order splitting method (Karniadakis et al., 1991) and second-order accurate time stepping is used in the current study.

The self-excited lift force at instantaneous timestep is obtained by integrating the pressure distribution on the bridge deck surface in the noninertial coordinate system, since the pressure is same in these two sets of coordinate system.

2.3. Extraction of aerodynamic damping and estimation of steady-state VIV amplitude

For the bridge deck undergoing forced vibration $h(t) = A_h \cos(\omega t)$ in the vertical direction, the simulated self-excited lift force $L_{se}(t)$ can be expressed in terms of the aerodynamic derivatives (Chen et al., 2000; Chen and Kareem, 2002; Scanlan, 1978):

$$L_{se}(t) = \frac{1}{2} \rho U^2 (2b) \left(k H_1^* \frac{\dot{h}}{U} + k^2 H_4^* \frac{h}{b} \right) \quad (6)$$

where $k = \omega b/U$ is the reduced frequency; $B = 2b$ is the bridge deck width; $\omega = 2\pi f$ is the circular frequency of the vibration; H_1^* and H_4^* are aerodynamic derivatives, which can be estimated from the simulated self-excited lift force coefficient $C_{Lse}(t) = L_{se}(t)/(\rho U^2 b)$ by harmonic balance approach:

$$k^2 H_1^* = -\frac{\omega b}{\pi A_h} \int_0^{2\pi/\omega} C_{Lse}(t) \sin(\omega t) dt \quad (7a)$$

$$k^2 H_4^* = \frac{\omega b}{\pi A_h} \int_0^{2\pi/\omega} C_{Lse}(t) \cos(\omega t) dt \quad (7b)$$

At the vicinity of the vortex lock-in region, H_1^* and H_4^* are functions of both reduced frequency and nondimensional vibration amplitude $A = A_h/D$, where D is the bridge deck depth, i.e., $H_1^* = H_1^*(k, A)$ and $H_4^* = H_4^*(k, A)$ (e.g., Chen, 2013).

The aerodynamic derivatives H_1^* and H_4^* can be used to predict the vertical VIV of the bridge deck by solving the following equation of motion:

$$m \left(\ddot{h} + 2\xi_{hs} \omega_{hs} \dot{h} + \omega_{hs}^2 h \right) = \frac{1}{2} \rho U^2 (2b) \left(k H_1^* \frac{\dot{h}}{U} + k^2 H_4^* \frac{h}{b} \right) \quad (8)$$

where ω_{hs} and ξ_{hs} are frequency and damping ratio of the bridge deck; m is bridge deck mass per unit spanwise length. For vertical VIV of bridge deck, the influence of aerodynamic stiffness due to H_4^* on structural frequency is generally small and can be neglected. From Eq. (8), the total system damping including both structural and aerodynamic damping is given by

$$\xi_h = \xi_{hs} + \xi_{ha} = \xi_{hs} - \frac{1}{2} \left(\frac{\rho b^2}{m} \right) H_1^* \quad (9)$$

where ξ_{ha} is the aerodynamic damping ratio.

The VIV occurs when the total system damping is negative. The VIV reaches a steady-state amplitude when the total system damping becomes zero, which is

$$H_1^* - \frac{2}{\pi} S_c = 0 \quad (10)$$

where $S_c = 4\pi m \xi_{hs} / (\rho B^2)$ is the Scruton number. Eq. (10) is used to determine the VIV amplitude at given wind speed and structural parameters. It is evident that the influence of structural mass and damping parameters on VIV amplitude can be unified by the Scruton number.

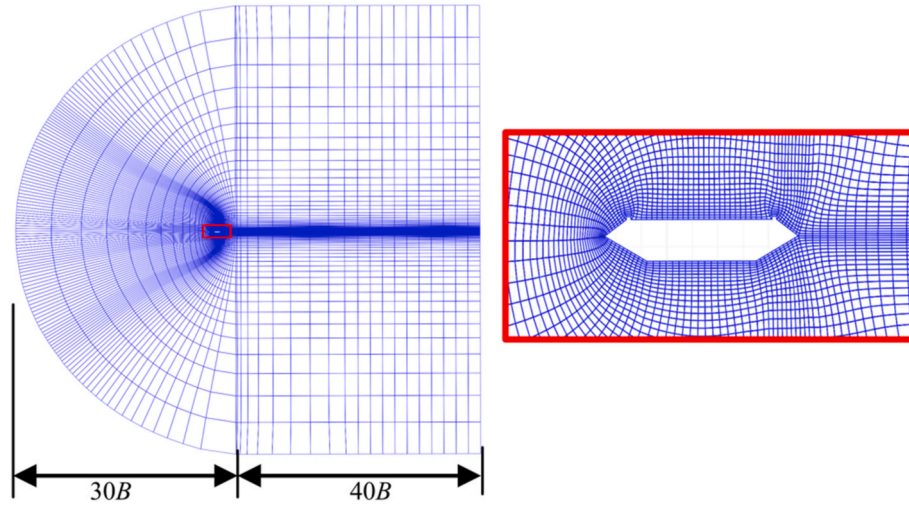


Fig. 3. Computational domain and C-grid around bridge deck.

3. Simulation of VIV of a streamlined box deck section

3.1. Experimental and simulation setup

The bridge deck section studied by [Noguchi et al. \(2020\)](#) for VIV response with wind tunnel test in smooth flow (turbulence intensities were about 1%) is used for current numerical study. The bridge deck cross-section is shown in [Fig. 2](#). The depth D , width B , and length L of the section model used in the experiment were 30 mm, 165 mm, and 1000 mm, respectively, i.e., the side ratio of the cross-section $B/D = 5.5$. The mass per unit length was $m = 2.016$ kg/m. The bridge deck was spring-supported in vertical direction with natural frequency $f_{hs} = 3.104$ Hz and structural damping ratio ξ_{hs} , which was found to be affected by vibration amplitude, i.e., $\xi_{hs} = (0.0213 A_h/D + 0.002)/(2\pi)$. The steady-state vertical VIV amplitudes at various reduced velocities were obtained, which will be used to validate the accuracy of current simulation.

The computational domain and C-grid around bridge deck are shown in [Fig. 3](#). The mesh around the bridge deck was generated through hyperbolic extrusion algorithm to form the half-circle block. The radius of the half-circle block is $30B$ and the width of the rectangular block is $40B$. The simulation was conducted by specifying normalized parameters (depth is scaled to be $D = 1$) and the actual unit of computational domain is not important. The thickness of first layer was chosen according to Reynolds number of 1000 (close to Reynolds number of target wind tunnel test) and $y^+ = y_G u_G / \nu \approx 1$ where G is the first layer grid near the wall, y_G and u_G are distance from the wall and streamwise velocity at grid G , respectively. The expansion ratio of the grid starting from the wall was chosen as 1.1. The total number of 8500 quadrilateral elements in the C-grid were used to discretize the computational domain in the $(\hat{x}, \hat{y})$ -plane. For the baseline simulations, 5th order polynomial expansion functions were used inside each quadrilateral element; and the spanwise size of the domain was chosen to be $L_z/D = \pi$ with total 16

Fourier modes to represent the solutions in z -direction. The small value of L_z/D was selected to ensure that the assumption of section model is valid since 3D flow structure will be evident for large spanwise size. The open-source spectral element framework, Nektar++ ([Moxey et al., 2020](#)), was used in current study.

The lift force time history was simulated at different reduced velocity U/fD (in the following U/fD is used rather than U/fB as it is often used for VIV of bluff bodies) and nondimensional amplitude A_h/D , where U/fD was set from 4 to 13 with an increment of 1; and A_h/D from 0.025 to 0.2 with an increment of 0.025. Local refinement of U/fD was added at 4.5, 5.5, 11.5, and 12.5. The imposed vibration frequency f was fixed and U/fD was changed by varying incoming wind velocity U . The Reynolds number in simulations was chosen to be 1000, which is close to the Reynolds number used in the target wind tunnel tests (in the range from 700 to 1800 when reduced velocity changes from 4 to 10). The fluid dynamic viscosity used in the simulation was changed accordingly to maintain a constant Reynolds number of 1000.

It is observed that the transient effects of initial condition to the simulated self-excited lift force time history can be eliminated by discarding the first 5 cycles of data. The total number of simulated cycles was extended for some cases for the estimation of power spectral density (PSD) function of self-excited lift time history. The time increment was set to be $f\Delta t = 5.0 \times 10^{-5}$ in the simulations to maintain the Courant-Friedrichs-Lewy (CFL) number less than 1. The simulations were conducted in the high-performance computing cluster at Texas Tech University. It required about 92 h to finish a single case using parallel computing with 144 processes (Intel Xeon E5-2695 v4 processors).

Convergence analysis was conducted regarding the influence of Lagrange polynomial order, number of Fourier modes, spanwise size and cycles (length) of the simulated data on the target indicators, i.e., the standard deviation (STD) of lift force coefficient $C_{Lse}(t)$, and aerodynamic derivatives H_1^* and H_4^* . The results for $A_h/D = 0.1$ and $U/fD = 10$ are shown in [Table 1](#). Comparing the results of Case 1 and 3 with 6, it

Table 1

Convergence analysis of forced vibration simulation ($A_h/D = 0.1$, $U/fD = 10$).

Case ID	Polynomial Order	Fourier mode number	Spanwise size L_z/D	Cycle number	C_{Lse} (STD)	H_1^*	H_4^*
1	3	16	π	10	0.213	0.768	2.621
2	5	8	π	10	0.211	0.878	2.479
3	5	16	π	10	0.207	0.732	2.537
4	5	16	π	50	0.207	0.756	2.544
5	5	32	2π	10	0.208	0.730	2.56
6	7	16	π	10	0.205	0.724	2.507
7	5	32	π	10	0.208	0.736	2.528
8	7	32	π	10	0.207	0.735	2.517

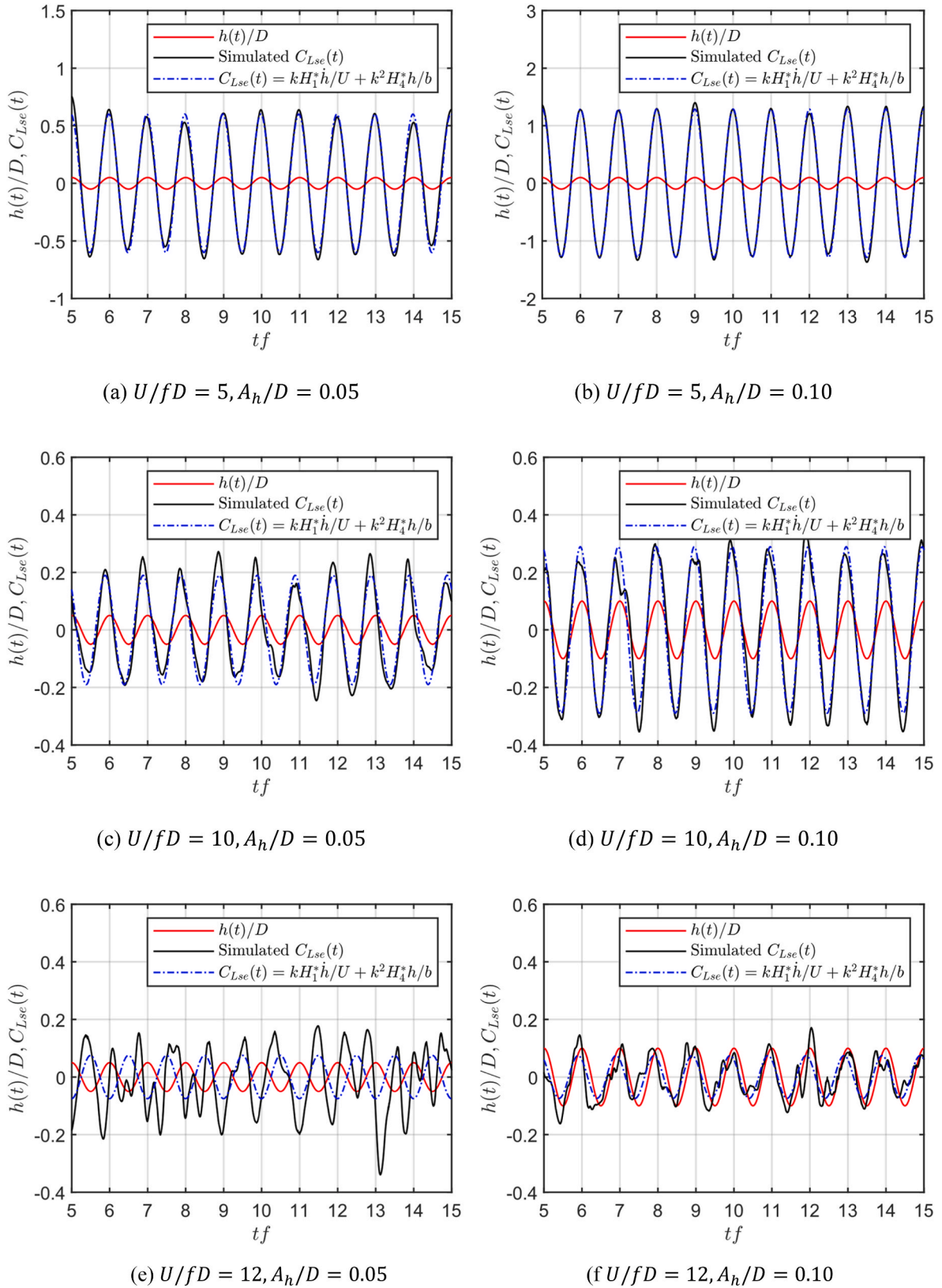


Fig. 4. Time history samples of deck displacement $h(t)/D$, simulated self-excited lift force coefficient $C_{Lse}(t)$ and its representation by aerodynamic derivatives.

is observed that the predicted H_1^* with the polynomial order of 3 (Case 1) and 5 (Case 2) has a relative error of 6.1% and 1.1%, respectively, compared to polynomial order of 7 (Case 6). The polynomial order of 5 is regarded to generate convergent prediction. Similar analysis for

increased number of Fourier modes in Cases 2, 3, and 7 shows that Fourier modes of 16 can generate convergent prediction. By comparing Cases 3 and 5, the spanwise size $L_z/D = \pi$ is enough for spanwise size compared with larger L_z/D . By comparing Cases 3 and 4, estimation of

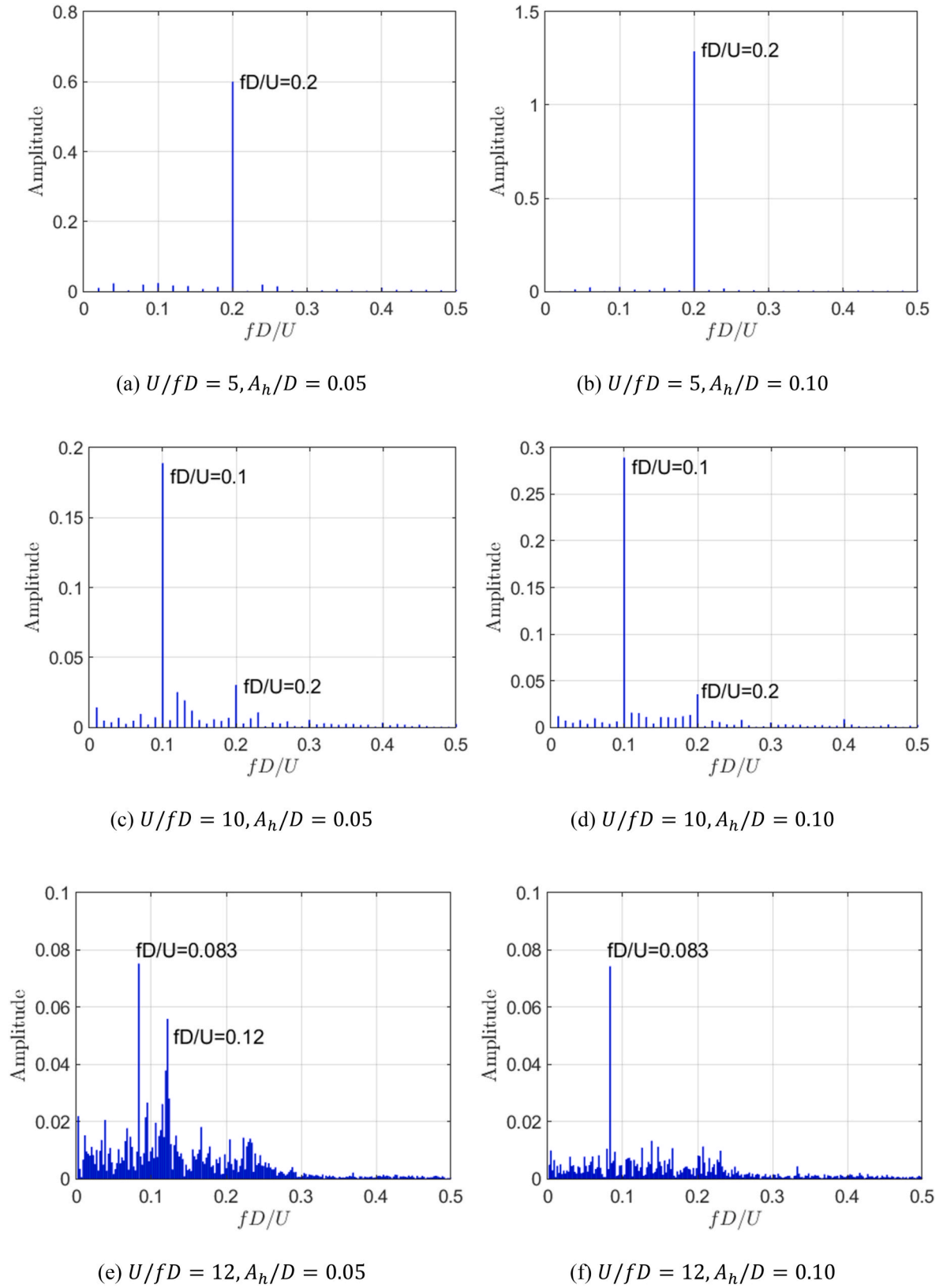


Fig. 5. Amplitude spectra of self-excited lift force coefficient. $C_{Lse}(t)$.

aerodynamic derivatives by 10 cycles of data is accurate enough compared with 50 cycles.

3.2. Aerodynamic damping and steady-state VIV amplitude

Fig. 4 shows time history samples of deck displacement $h(t)/D$, self-excited lift force coefficient $C_{Lse}(t)$ and its representation by

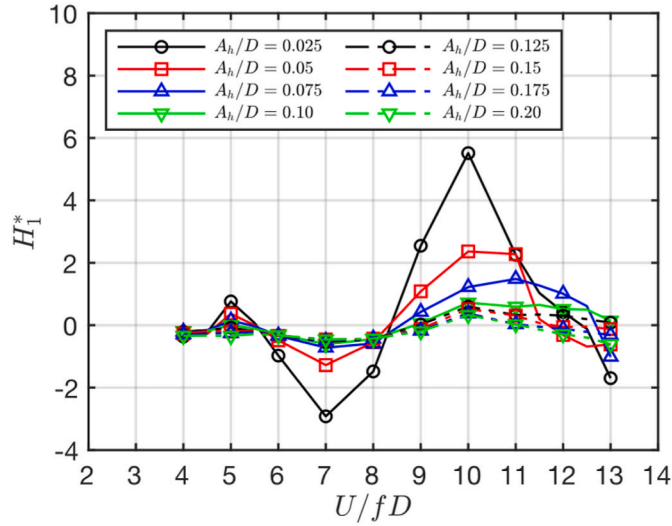


Fig. 6. Aerodynamic derivative H_1^* as a function of U/fD and A_h/D .

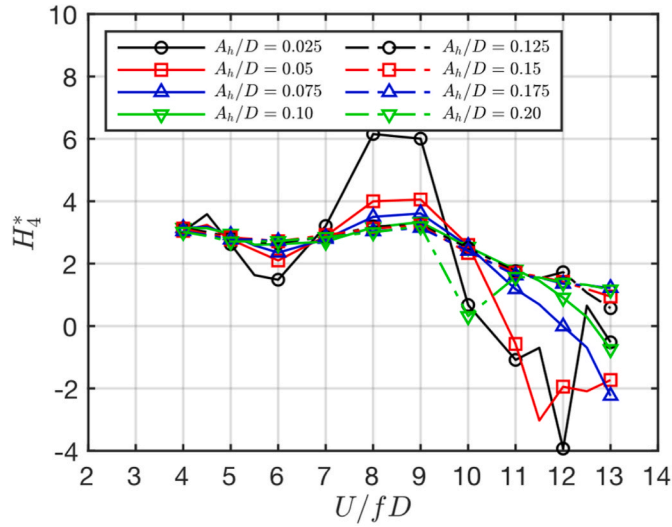
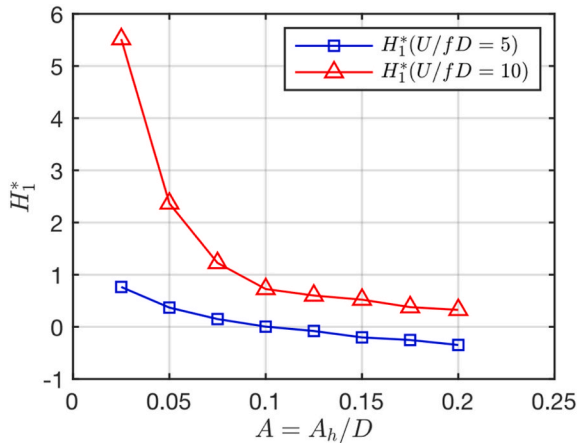
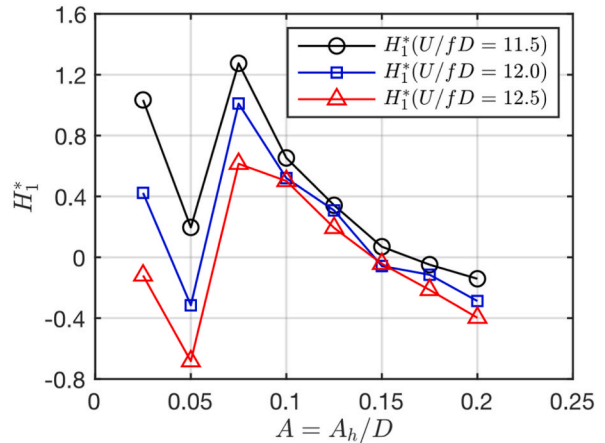


Fig. 7. Aerodynamic derivative H_4^* as a function of U/fD and A_h/D .



(a) Monotonic aerodynamic damping



(b) Non-monotonic aerodynamic damping

Fig. 8. Amplitude-dependent aerodynamic damping for given reduced wind velocity.

aerodynamic derivatives, $kH_1^* \dot{h}/U + k^2 H_4^* h/b$, at reduced wind speed $U/fD = 5, 10, 12$ and $A_h/D = 0.05, 0.10$. The amplitude spectra of $C_{Lse}(t)$ from these cases are shown in Fig. 5. For cases at $U/fD = 5$ and 10 , the dominant frequency is corresponding to the forced vibration frequency, showing “lock-in” phenomenon. It is worth noting that at $U/fD = 10$, there exists low-level of higher-order harmonic component as seen in Fig. 5(c and d), which is also responsible to slight double peaks in corresponding time histories shown in Fig. 4(c and d). For the case at $U/fD = 12, A_h/D = 0.05$, two dominant frequencies corresponding to $fD = 8.33$ and 12 ($fD/U = 0.12$ and 0.083) are observed in Fig. 5(e), which indicates that perfect “lock-in” is no longer satisfied and desynchronization occurs. With increased amplitude at $U/fD = 12, A_h/D = 0.10$, “lock-in” reappears as seen in Fig. 5(f). It should be mentioned that higher-order harmonic components are orthogonal with the structural vibration harmonic, thus have no contribution to the aerodynamic damping. For the prediction of VIV response, the modeling of aerodynamic damping is critical rather than reconstructing the time history of actual self-excited lift force. For the case at $U/fD = 12, A_h/D = 0.05$, the computed aerodynamic damping includes contributions from both frequency components.

The estimated H_1^* and H_4^* by Eq. (7) are plotted in Fig. 6 and Fig. 7, where the amplitude dependence is evident. It is observed that there are two possible VIV regions near $U/fD = 5$ and 10 , where the aerodynamic damping (aerodynamic derivative H_1^*) is negative (positive) at lower amplitude of vibration.

The steady-state amplitude of VIV at a specified reduced velocity for a given level of structural damping ratio ξ_{hs} , or Scruton number, can be predicated by solving the equation of zero total system damping (including structural damping and aerodynamic damping) in Eq. (10). When the aerodynamic damping monotonically varies with vibration amplitude as shown in Fig. 8(a), a single steady-state amplitude of VIV can be determined. At $U/fD = 5$, the steady-state VIV amplitude is predicted to be $A_h/D = 0.043$ for $\xi_{hs} = 0.1\%$ ($S_C = 0.76$) as seen in Fig. 9(a); while VIV is not observed for $\xi_{hs} = 0.5\%$ ($S_C = 3.8$). At $U/fD = 10$, $\xi_{hs} = 0.1\%, 0.5\%$ and 1% ($S_C = 7.6$) correspond to $A_h/D = 0.157, 0.050$ and 0.030 , respectively.

The aerodynamic damping does not vary monotonically with vibration amplitude as shown in Fig. 8(b) around $U/fD = 12$ where the hysteresis phenomenon was observed. For instance, at $U/fD = 12$ under $\xi_{hs} = 0.05\%$ ($S_C = 0.38$) as shown in Fig. 9(b), there exist three amplitude levels, $A_h/D = 0.031, 0.061$, and 0.130 , which correspond to zero system damping. When the initial vibration amplitude $A_h/D < 0.031$, the total system damping is negative and the amplitude grows until reaching $A_h/D = 0.031$; When initial vibration amplitude

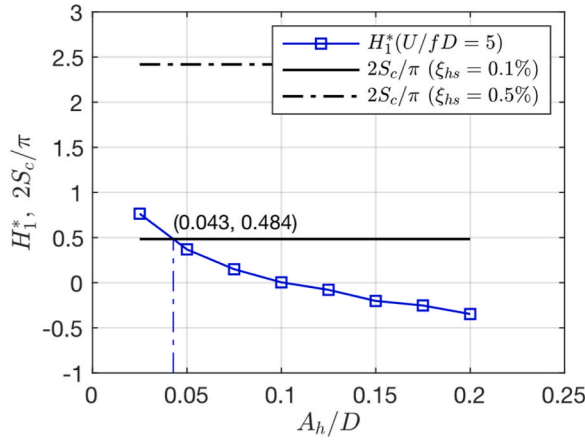
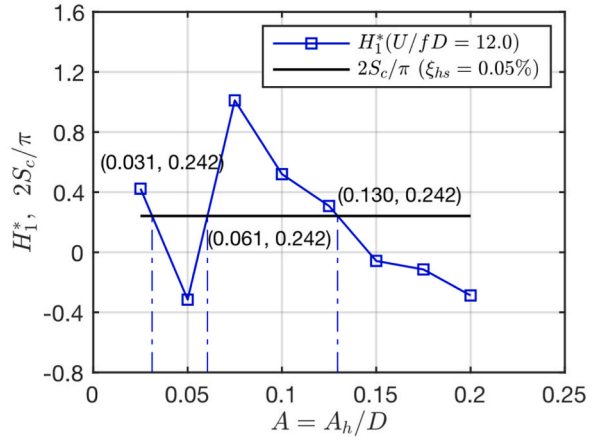
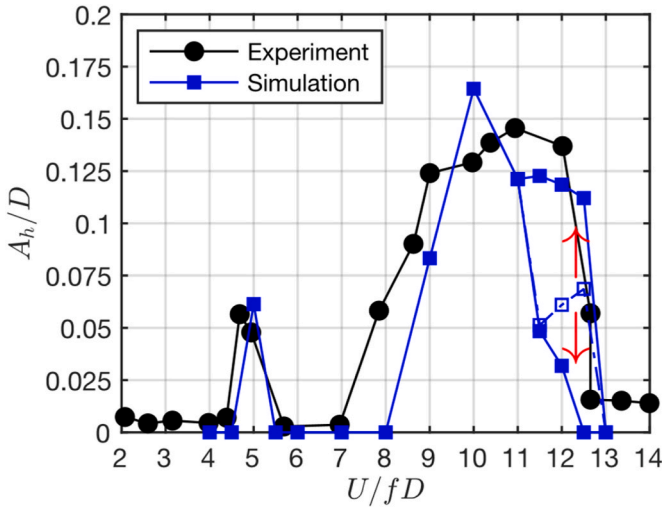
(a) $U/fD = 5$ (b) $U/fD = 12$

Fig. 9. Determination of VIV amplitude at given Scruton number.

Fig. 10. Comparison of predicted VIV amplitudes by current simulation and wind tunnel tests by [Noguchi et al. \(2020\)](#).

$0.031 < A_h/D < 0.061$, the total system damping is positive and the amplitude decays until reaching $A_h/D = 0.031$; When the initial vibration amplitude $0.061 < A_h/D < 0.130$, the total system damping is negative and the amplitude grows until reaching $A_h/D = 0.130$; When the initial vibration amplitude $A_h/D > 0.130$, the total system damping is positive and the amplitude decays until reaching $A_h/D = 0.130$. Clearly, there are two stable steady-state amplitudes, i.e., $A_h/D = 0.031$ and 0.130 depending on the initial condition. The non-monotonically varying aerodynamic damping with vibration amplitude results in a hysteresis phenomenon. The hysteresis phenomenon was also found in wind tunnel study of this section at about $U/fD = 12.6$ ([Noguchi et al., 2020](#)), and other bluff bodies ([Feng, 1968](#); [Morse and Williamson, 2009](#)). This hysteresis phenomenon will be further discussed based on the flow visualization of vortex shedding modes in Section 3.4.

The steady-state VIV amplitudes at various U/fD and $\xi_{hs} = (0.0213 A_h/D + 0.0022)/(2\pi)$ are calculated and compared with wind tunnel test results ([Noguchi et al., 2020](#)) as shown in Fig. 10. The VIV wind velocity regions and amplitudes are predicted by current simulation with good accuracy. It is worth noting that two possible stable amplitude branches (solid line with square markers) exist in the hysteresis region for the simulation results, where the dash-dotted line denotes unstable amplitude solution. It can be interpreted in the same

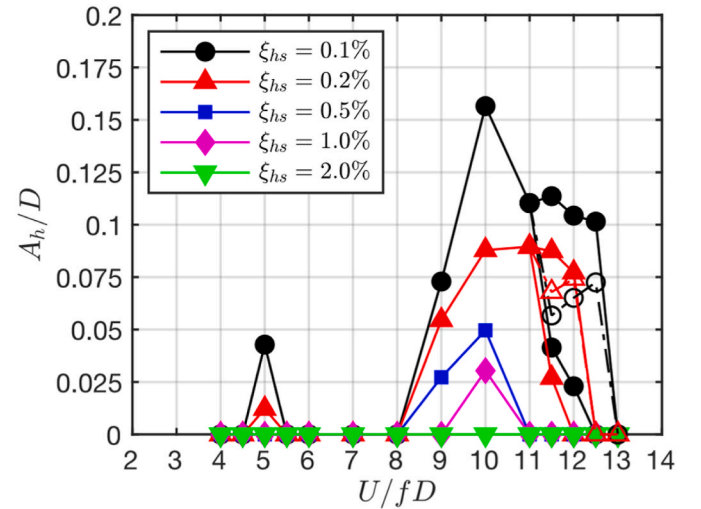


Fig. 11. Prediction of VIV amplitude of bridge deck with constant structural damping ratio.

manner as in Fig. 9(b) that the stable amplitude will be determined by the lower (upper) branch when the initial amplitude is smaller (larger) than the unstable amplitude solution.

Fig. 11 portrays the steady-state VIV amplitude as a function of wind speed and constant structural damping ratio. It is evident that increasing structural damping, or Scruton number, leads to decrease in the VIV amplitude and velocity region of VIV occurrence.

3.3. VIV generation mechanism

The generation mechanisms of VIV at reduced velocity regions near 5 and 10 will be discussed based on flow visualization of vortex shedding modes, distributions of dynamic pressure and induced energy transfer from flow to bridge deck.

The contour plots of pressure coefficient $C_p = p/(0.5\rho U^2)$ around the vibrating deck, at the time instant when deck reaches the maximum positive displacement of $0.05D$, are shown in Fig. 12, where the negative pressure peak indicates the existence of vortex cores. The local negative pressure peak is a simple criterion for vortex identification, and the pressure distribution is of most interest since it directly contributes to self-excited force. It is worth noting that other vortex identification methods exist based on velocity gradient such as Q and λ_2 criteria (e.g.,

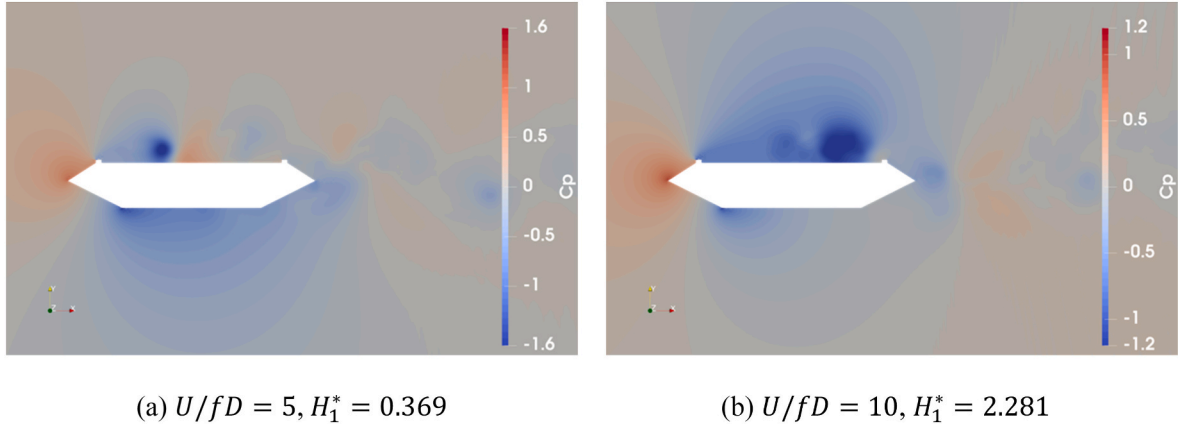


Fig. 12. Contour plots of pressure coefficient around the vibrating deck at the position of maximum positive displacement of 0.05D.

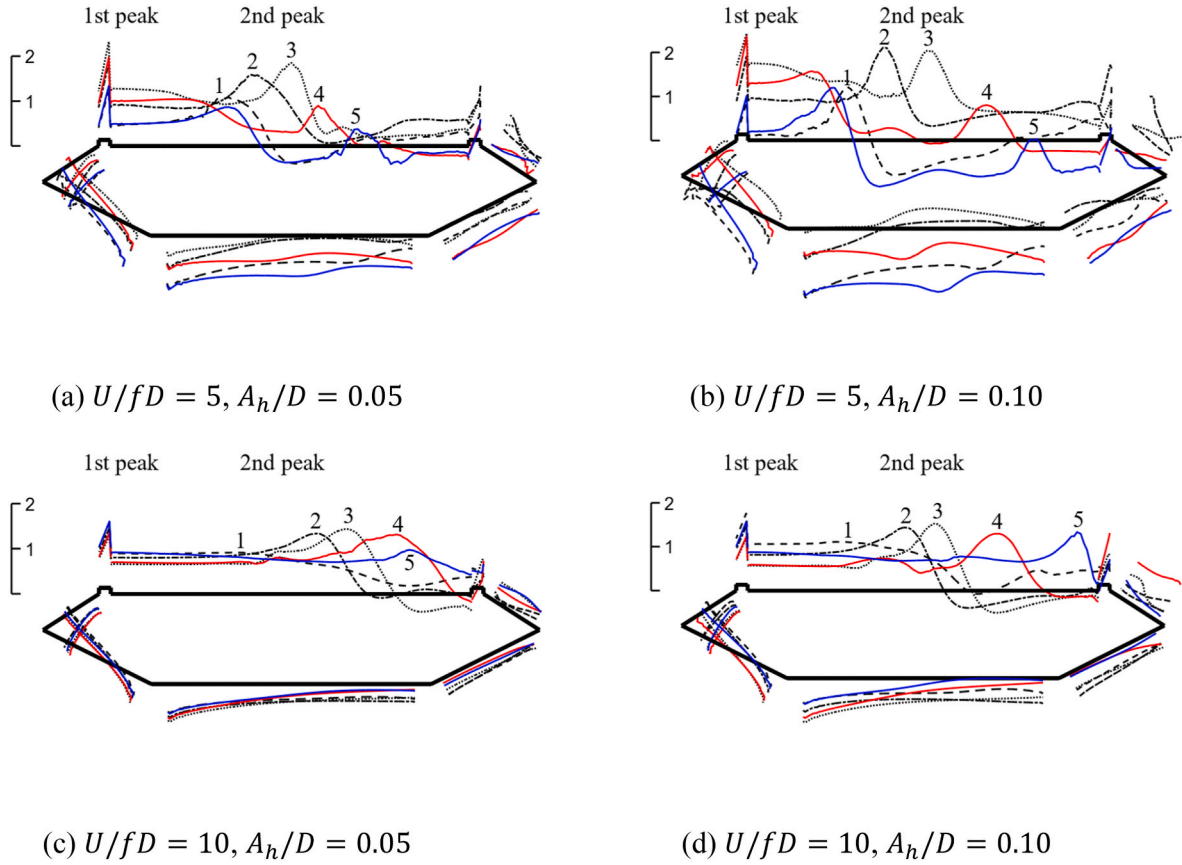


Fig. 13. Dynamic pressure coefficient distributions on the vibrating bridge decks at time instants of $2\pi ft = 0.4\pi$ (dashed line), 0.8π (dash-dotted line), 1.2π (dotted line), 1.6π (red solid line), 2.0π (blue solid line) during a vibration cycle, which are also numbered from 1 to 5. The outward lines indicate negative pressure while inward lines indicate positive pressure. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Jeong and Hussain, 1995), which are not used here since the velocity gradient field from spectral element simulation is not continuous across the element boundary, which will introduce much noise to the contour plot. It is evident that the flow is dominated by leading-edge vortex shedding, while the pattern of Karman vortex shedding is hardly recognized. This can also be understood from the distribution of dynamic pressures on the deck surface during a cycle of vibration shown in Fig. 13, where the first peak on the top left corner is the local separation bubble and the second wide peak is related to the leading-edge vortex. The rolling downstream of the leading-edge vortex is clearly seen from the temporal change of the second pressure peak locations. The

unphysical pressure oscillations in the small near-corner region as seen in Fig. 13 are related to the sharp corner singularity of numerical simulation.

From the dynamic pressure coefficient distribution $C_p(s, t) = p(s, t)/(0.5\rho U^2)$, where $p(s, t)$ is the dynamic pressure and s denotes the location on the surface of the bridge deck, the contribution of the pressures to the aerodynamic derivative H_1^* (or aerodynamic damping), $h_1^*(s)$, can be calculated as follows:

$$C_{Lse}(t) = \oint C_p(s, t) n_y(s) ds / (2b) \quad (11a)$$

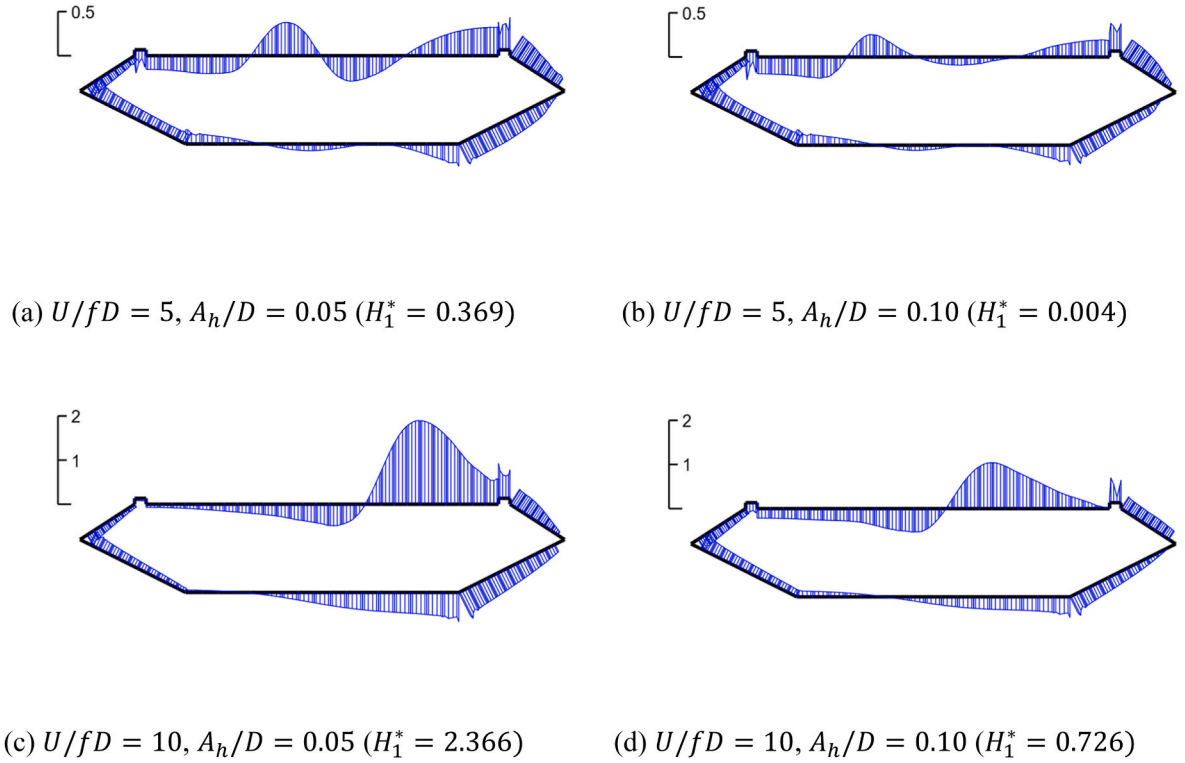


Fig. 14. $h_1^*(s)$ distribution on the bridge deck. The outward lines indicate negative aerodynamic damping (excitation) while inward lines indicate positive aerodynamic damping.

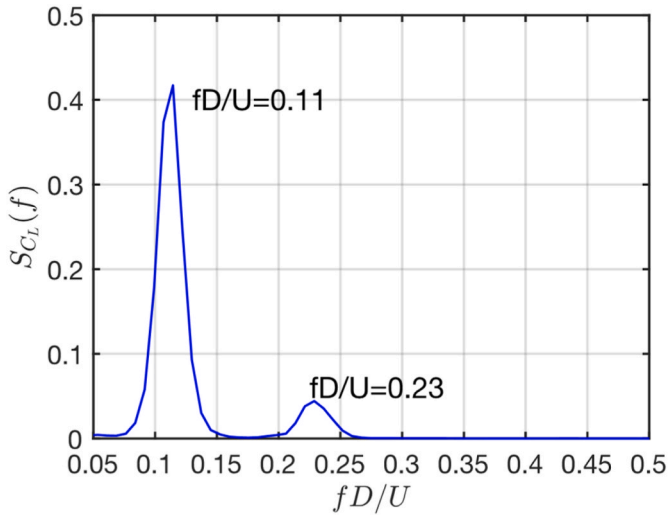


Fig. 15. PSD of lift force time history on the stationary bridge deck.

$$k^2 H_1^* = \oint k^2 h_1^*(s) ds \quad (11b)$$

$$k^2 h_1^*(s) = -\frac{\omega}{2\pi A_h} \int_0^{\frac{2\pi}{\omega}} C_p(s, t) n_y(s) \sin(\omega t) dt \quad (11c)$$

where $n_y(s)$ is the cosine of the angle between inward normal vector and vertical (lift) direction. $h_1^*(s)$ denotes the contribution of pressure per unit length at s location to aerodynamic derivative H_1^* . A positive value of $h_1^*(s)$ indicates a negative aerodynamic damping and excitation effect.

The $h_1^*(s)$ distributions on the bridge deck at $U/fD = 5, 10$ and $A_h/D = 0.05, 0.10$ are plotted in Fig. 14. It can be seen that $h_1^*(s)$ distributions are quite different in two VIV ranges, which is due to the difference in the number of separated vortices from the leading edge to reach the trailing edge in each vibration cycle.

According to Shiraishi and Matsumoto (1983), the moving speed of the separated vortices of rectangular cylinders is around 60% of free stream wind speed. The time of vortex traveling from leading edge to trailing edge is around $B/(0.6U)$. Denoting n as the number of vortices separated from leading edge during a vibration cycle, the resulted lift frequency is $0.6nU/B$. The VIV occurs when the lift frequency is close to the vibration frequency of the body (f), which leads to the critical reduced velocity for VIV as $U/fD = 1.67 (1/n)(B/D)$. The critical reduced velocities evaluated for $B/D = 5.5$, $n = 1$ and $n = 2$ are 9.2 and 4.6 respectively, which are in good agreement with the prediction (Fig. 10) even for current non-rectangular bridge deck. On the other hand, the reduced frequency of Karman vortex shedding, i.e., Strouhal number St , can be determined from the PSD of lift force on the stationary bridge deck (Fig. 15), where $fD/U = 0.23$ is the Strouhal number, since it is close to that of the same bridge deck without barriers (0.231–0.242), while $fD/U = 0.11$ is the reduced frequency of the stronger leading-edge vortex shedding. It is worth noting that the reduced velocities from $1/St = 4.3$ and $1.67 (1/n)(B/D) = 4.6$ with $n = 2$ have rather close values for current deck section. However, based on the observations in Figs. 12, Figs. 13 and 14 (also discussions in Section 3.5 on Karman vortex shedding from bridge deck without barriers), it can be inferred that the shedding of leading-edge vortex rather than Karman vortex is the dominant mode contributing to the VIV of current bridge deck in both reduced velocity regions around $U/fD = 5$ and 10. It is worth noting that the Strouhal number, as the characteristic frequency of Karman vortex shedding, is not directly related to LEV and VIV performance of current bridge deck with barriers.

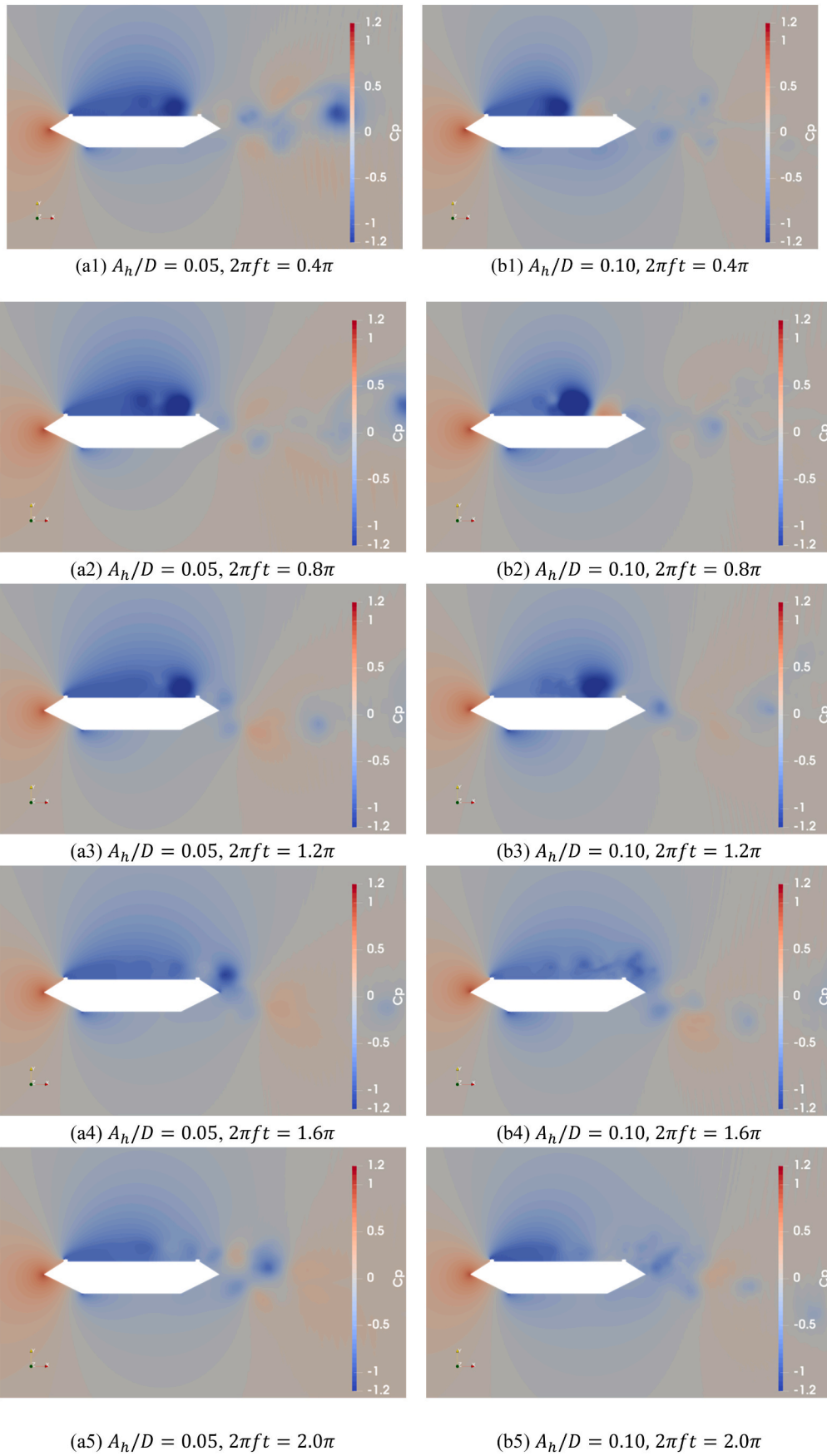
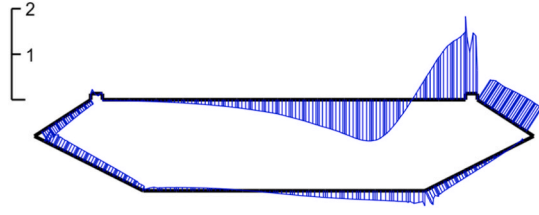
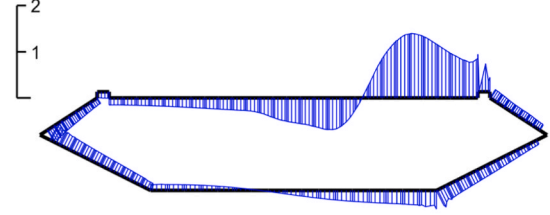
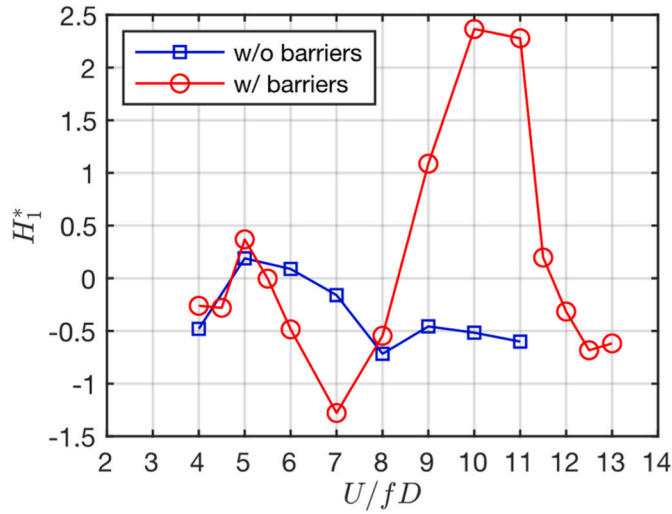


Fig. 16. Contour plots of pressure coefficient in the wake of vibrating deck during a vibration cycle at $U/fD = 12$.

(a) $A_h/D = 0.05$, $H_1^* = -0.315$ (b) $A_h/D = 0.10$, $H_1^* = 0.520$ Fig. 17. $h_1^*(s)$ distribution on the bridge deck at $U/fD = 12$.Fig. 18. Aerodynamic derivative H_1^* for the bridge deck with and without barriers at $A_h/D = 0.05$.

3.4. Mechanism of hysteresis phenomenon

Fig. 16 shows the contour plots of pressure coefficient around the vibrating deck section during a vibration cycle $h(t) = A_h \cos(2\pi ft)$ at time instants of $2\pi ft = 0.4\pi, 0.8\pi, 1.2\pi, 1.6\pi$, and 2.0π under $U/fD = 12$, and $A_h/D = 0.05$ and 0.10 . It can be seen that the flow at lower vibration amplitude $A_h/D = 0.05$ is characterized by significant impinging of leading-edge vortex, where the separated leading-edge vortex escapes past the trailing edge, and also induces a secondary vortex on the opposite site (Nakamura and Nakashima, 1986), resulting to additional vortex shedding frequency shown in Fig. 5(a). The wake flow pattern after the body looks like Karman vortex street, but these two vortex

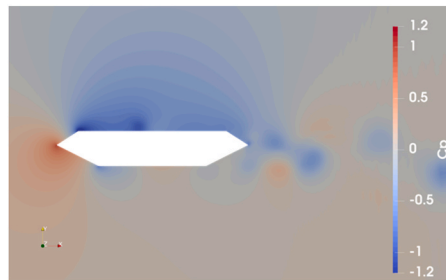
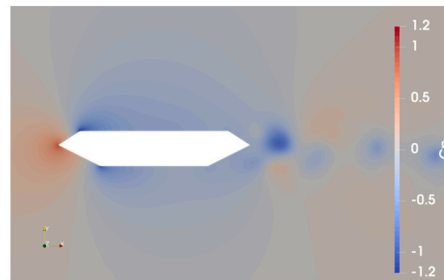
patterns should be differentiated. The flow structures at higher amplitude $A_h/D = 0.10$, on the other hand, is dominated by leading-edge vortex shedding. The hysteresis phenomenon observed near $U/fD = 12$ may be due to the influence of the impinging leading-edge vortex. It is worth noting that the mechanisms of hysteresis phenomenon of bridge deck are different from that of circular cylinders. The hysteresis phenomenon of circular cylinders is due to intermittent change of Karman vortex shedding modes, e.g., single vortex shedding mode and pair vortex shedding mode (Morse and Williamson, 2009).

The $h_1^*(s)$ distributions at $U/fD = 12$, $A_h/D = 0.05$ and 0.10 are shown in Fig. 17. Comparing the two cases and the case with $U/fD = 10$, the impinging of leading-edge vortex primarily affects the size of upper excitation region with positive $h_1^*(s)$ where the vibrating body extracts energy from the flow, leading to the change in H_1^* and aerodynamic damping.

3.5. Influence of deck barriers

The bridge deck section discussed above has both barriers (protrusions) near the leading and trailing edges as shown in Fig. 2. The role of barriers is also investigated by comparing the aerodynamics of bridge section with and without barriers. Fig. 18 displays the comparison of H_1^* at $A_h/D = 0.05$ and different reduced velocities. For the bridge section without barriers, at $U/fD = 5$, $H_1^* = 0.189$ that corresponds to a negative aerodynamic damping, indicating possibility for VIV to occur. However, it is lower than that of deck with barriers. Around $U/fD = 10$, local reduction of aerodynamic damping is observed but it remains positive.

The contour plots of pressure coefficient around the vibrating deck without barriers at the position of maximum positive displacement of $0.05D$, $U/fD = 5$ and 10 are shown in Fig. 19. Fig. 20 displays the dynamic pressure coefficients on the deck surface during a cycle of vibration. At $U/fD = 5$, the flow shows shedding of both leading-edge vortex and Karman vortex (Fig. 19(a)), and the dynamic pressure distribution

(a) $U/fD = 5$, $H_1^* = 0.189$ (b) $U/fD = 10$, $H_1^* = -0.516$ Fig. 19. Contour plots of pressure coefficient around the vibrating deck without barriers at the position of maximum positive displacement of $0.05D$.

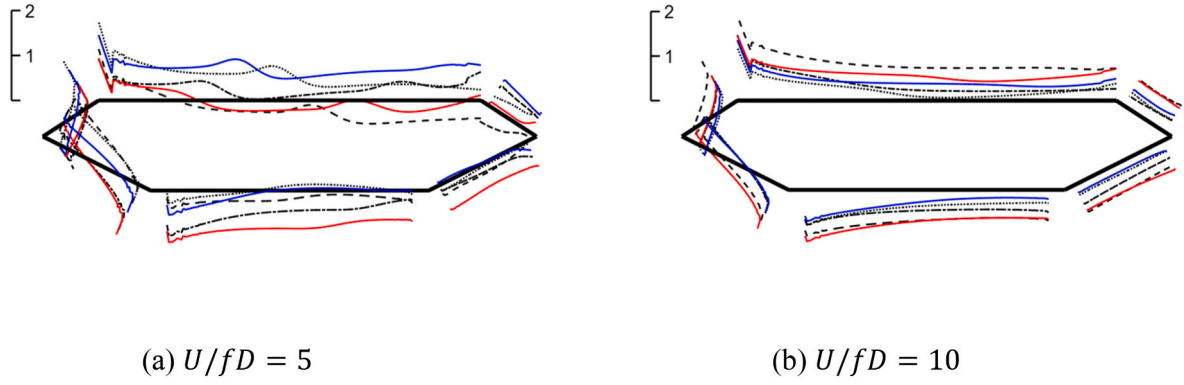


Fig. 20. Dynamic pressure coefficient distributions on the vibrating bridge decks without barriers at $A_h/D = 0.05$ and time instants of $2\pi ft = 0.4\pi$ (dashed line), 0.8π (dash-dotted line), 1.2π (dotted line), 1.6π (blue solid line), 2.0π (red solid line) during a vibration cycle. The outward lines indicate negative pressure while inward lines indicate positive pressure. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

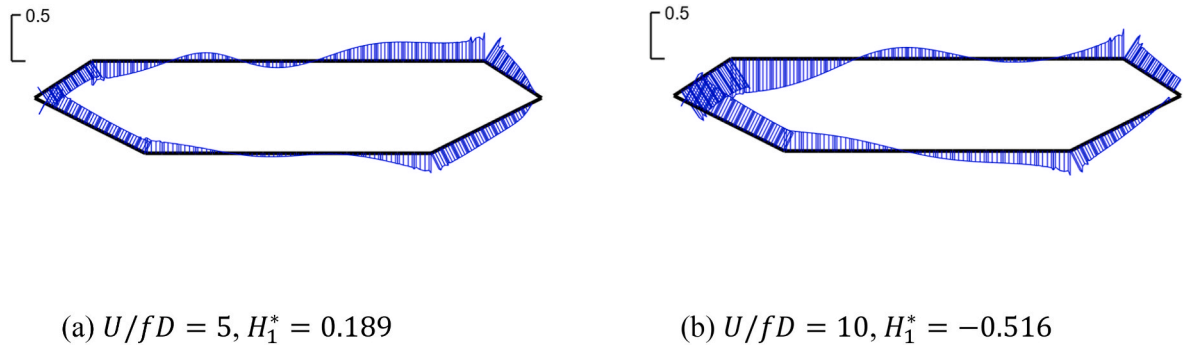


Fig. 21. $h_1^*(s)$ distribution on the bridge deck without barriers at $A_h/D = 0.05$.

(Fig. 20(a)) shows slight position change of pressure peak on upper deck surface related to rolling downstream of relatively weak leading-edge vortex. At $U/fD = 10$, the flow is dominated by Karman vortex shedding (Fig. 19(b)), and the temporal change of dynamic pressure distribution (Fig. 20(b)) does not show the characteristics related to rolling downstream of leading-edge vortex as seen in Fig. 13 for the deck with barriers. The $h_1^*(s)$ distributions on the bridge deck without barriers are also shown in Fig. 21 for comparison. The $h_1^*(s)$ distributions have similar patterns between $U/fD = 5$ and 10, which differs from the bridge deck with barriers (Fig. 14), indicating a change of vortex shedding

mode with the existence of barriers. It is worth noting that the relation between vortex shedding mode and $h_1^*(s)$ distribution is difficult to be quantified. Nevertheless, the $h_1^*(s)$ can provide some hints on possible suppression measures. These observations show that the leading barrier induces strong leading-edge vortex shedding and makes the bridge deck more susceptible to VIV.

4. Forced vibration with a varying amplitude

The standard forced vibration with sinusoidal displacement having a

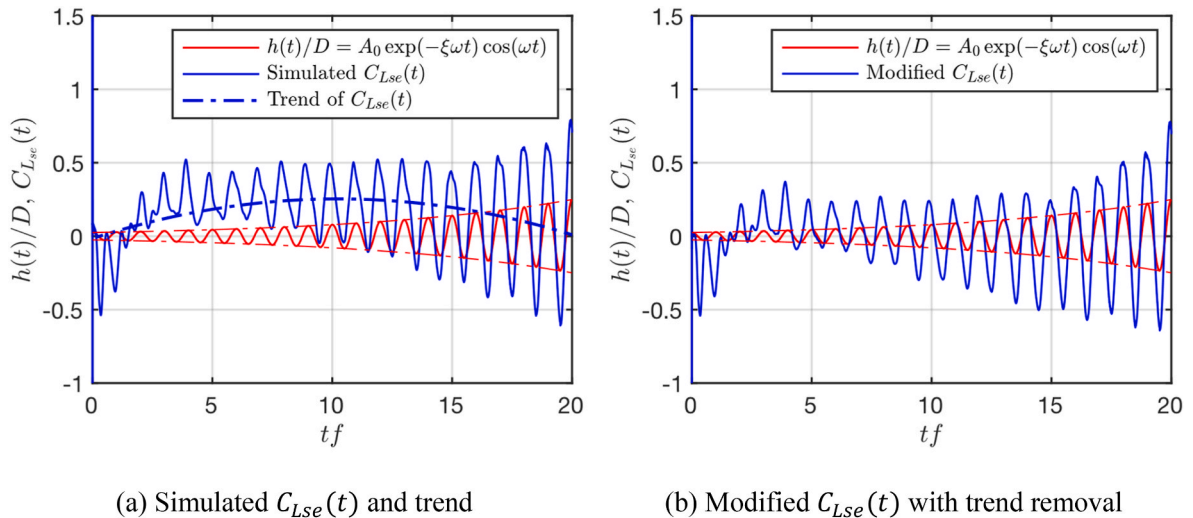


Fig. 22. Simulated lift force coefficient $C_{Lse}(t)$ from forced vibration with a varying amplitude at $U/fD = 10$.

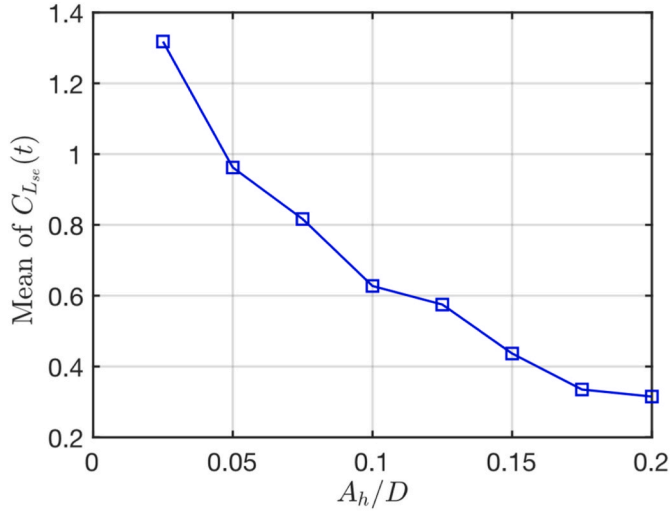


Fig. 23. Mean of $C_{Lse}(t)$ from standard forced vibration simulations at $U/fD = 10$.

constant amplitude can provide aerodynamic damping at one parameter pair of reduced velocity and amplitude U/fD . It requires large number of CFD runs to determine the aerodynamic damping for varied A_h/D and U/fD . To increase the computational efficiency, the forced vibration with a varying amplitude is explored for the estimation of amplitude-dependent aerodynamic damping at a specific reduced velocity with only one CFD run. Compared with free vibration method, the forced vibration with varying amplitude has better control of the interested amplitude range. From the perspective of numerical simulation, forced vibration with varying amplitude is easier to be implemented compared to free vibration since the former does not require the coupling between flow and structural solvers.

The modified sinusoidal vertical displacement of deck section has an exponentially increasing amplitude, which is corresponding to the free vibration of a single degree of freedom with a negative damping:

$$h(t) = A_0 \exp(-\xi \omega t) \cos(\omega t) \quad (12)$$

where A_0 is the initial amplitude; ξ is the damping ratio chosen to be negative in current case, which is related to the amplitude increasing rate.

It is noted that Siedziako and Øiseth, 2018, 2019) also conducted forced vibration experiments with varying amplitude to examine the validity of linear self-excited force model where the aerodynamic derivatives are considered to be independent of vibration amplitude. Current study proposes forced vibration with varying amplitude to extract amplitude-dependent aerodynamic damping efficiently, which is critical for the prediction of limit cycle amplitude of VIV.

The lift force coefficient time history $C_{Lse}(t)$ at $U/fD = 10$ with parameters of $A_0/D = 0.025$, $\xi = -0.0183$ is shown in Fig. 22(a). There exists a clear trend in the simulated time history, and an assumed quadratic trend is also shown in Fig. 22(a). The trend seems not to be related to the effect of initial conditions since it still exists when the flow is initialized using the results of another simulation rather than initialized by mean velocity field and imposed white noise. In the simulation of forced vibration with a constant amplitude, there is a mean of $C_{Lse}(t)$, which decreases with the increase in vibration amplitude as shown in Fig. 23 at $U/fD = 10$. The trend may be attributed to the change of mean of $C_{Lse}(t)$ with varying vibration amplitude. To the authors' knowledge, this observation has not been validated in experimental results, and it may be introduced by the current numerical simulation method. It should be mentioned that the existence of trend has negligible influences on the calculated aerodynamic derivatives through Eq. (7). The simulated $C_{Lse}(t)$ is modified with trend removal as shown in Fig. 22(b).

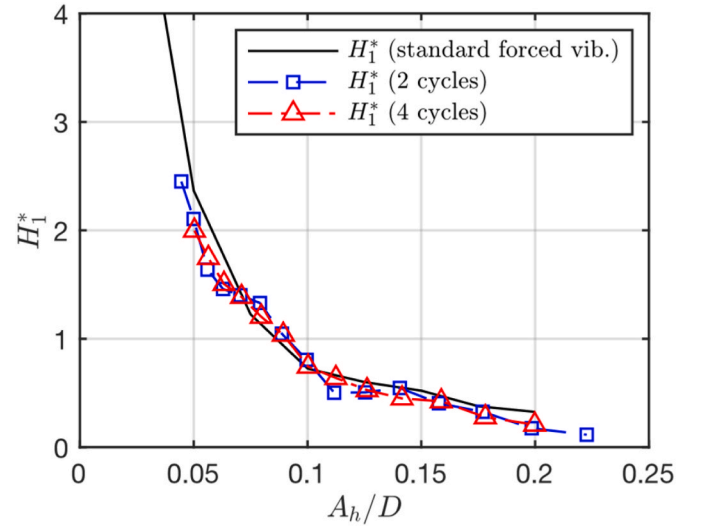


Fig. 24. Comparison of estimated aerodynamic derivatives H_1^* at $U/fD = 10$ from standard forced vibration and varying amplitude vibration.

Since the vibration amplitude only varies slowly with time, a simple method is used to estimate amplitude-dependent H_1^* from the lift time history. Several vibration cycles of simulated history of $C_{Lse}(t)$ (or modified one) is selected and H_1^* is estimated using previously described method (Eq. (7)). The mean amplitude of these cycles of deck vibration is chosen to be the corresponding amplitude.

The comparison of estimated H_1^* from standard forced vibration and varying amplitude vibration is shown in Fig. 24, which presents promising accuracy from varying amplitude simulation, especially when the average of 4 cycles is used.

It is shown in Fig. 25(a) that the parameter ξ , which corresponds to the amplitude increasing rate, has influences to the aerodynamic derivative especially at large amplitudes. When the amplitude increasing rate increases, the simple averaging method to extract H_1^* using mean amplitude during several cycles may introduce some errors when H_1^* is relatively sensitive to vibration amplitude. Choosing a small absolute value of ξ can resolve this issue, avoiding the necessity of a more advanced model to extract the amplitude-dependent aerodynamic damping. This is also evident at the reduced velocity of 5 when H_1^* is near zero and choosing a smaller value of ξ can improve the estimation of H_1^* (Fig. 25(b)). As expected, when hysteresis phenomenon exists, the dependence of aerodynamic damping with vibration amplitude is complicated, the proposed forced vibration scheme with varying amplitude will not provide satisfactory results (Fig. 26).

5. Conclusions

This study presented CFD simulation of a streamlined bridge deck under forced vibration for extracting aerodynamic damping and prediction of vertical VIV. The Navier-Stokes equations formulated in the noninertial coordinate system fixed on the vibrating body was solved with high-order spectral/Fourier method to bypass the requirement of mesh updating. The self-excited lift force time history was obtained by integrating the pressure distribution on the bridge deck from which the aerodynamic derivatives and aerodynamic damping were determined as functions of reduced wind velocity and vibration amplitude. The steady-state VIV amplitude under given wind velocity and structural parameters were calculated by letting the total system damping to be zero. The high prediction accuracy of the VIV amplitude compared with the wind tunnel free vibration tests demonstrated that the simulation framework is appropriate for the bridge deck VIV simulation.

The generation mechanisms of VIV and hysteresis phenomenon for

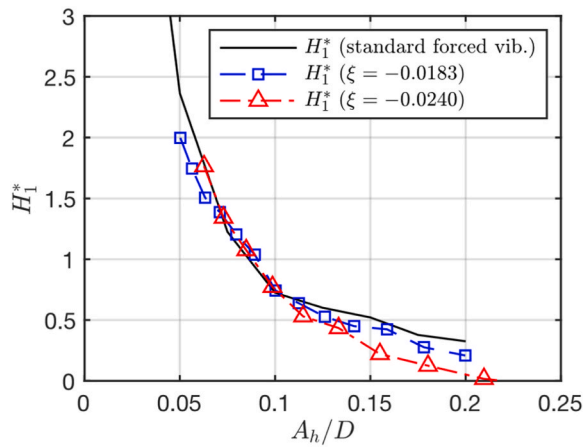
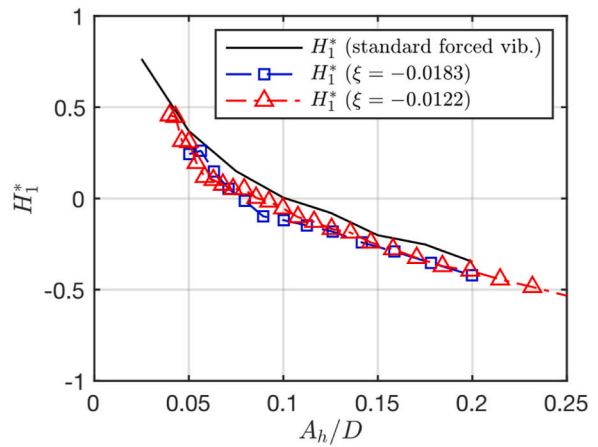
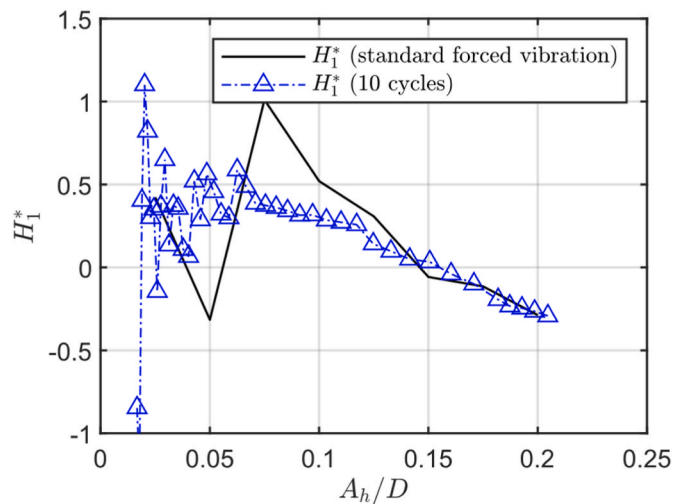
(a) $U/fD = 10$ (b) $U/fD = 5$

Fig. 25. Effects of amplitude growing rate to the estimation of aerodynamic derivative.

Fig. 26. Estimation of aerodynamic derivative at $U/fD = 12$ with $\xi = -0.01$.

current streamlined bridge deck were discussed based on flow visualization and dynamic pressure distribution. It was shown that at both VIV occurrence regions around the reduced velocities of 5 and 10, the VIV was caused by the shedding of leading-edge vortex rather than Karman vortex. The hysteresis phenomenon was evident from the non-monotonic relation between aerodynamic damping and vibration amplitude. The flow visualization also shows that the hysteresis is probably due to effect of impinging leading-edge vortex in addition to leading-edge vortex shedding, which has different mechanism from that of circular cylinders. The current bridge deck has two barriers which plays important role to the flow structure. The leading-edge vortex separates from the bridge deck with barriers, causing a large wake region behind the leading barrier. The same bridge deck without barriers showed lower negative aerodynamic damping and potential VIV only near reduced wind velocity of 5. From the dynamic pressure distributions on the surface of the vibrating deck, the contribution of pressures at different locations to aerodynamic damping was quantified, which can also provide some hints on VIV mechanisms and possible suppression measures.

The forced vibration with varying amplitude was explored to increase the computational efficiency, where the amplitude-dependent aerodynamic damping can be estimated from a single CFD run. It was shown that the modified amplitude method can estimate the amplitude-

dependent aerodynamic damping with good accuracy compared with standard forced vibration, except in the hysteresis region, where the non-monotonic property is more difficult to be captured.

Author declaration

[Instructions: Please check all applicable boxes and provide additional information as requested.]

Funding

No funding was received for this work.

Intellectual property

xWe confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

Research ethics

xWe further confirm that any aspect of the work covered in this manuscript that has involved human patients has been conducted with the ethical approval of all relevant bodies and that such approvals are acknowledged within the manuscript.

xIRB approval was obtained (required for studies and series of 3 or more cases)

Written consent to publish potentially identifying information, such as details or the case and photographs, was obtained from the patient(s) or their legal guardian(s).

Authorship

The International Committee of Medical Journal Editors (ICMJE) recommends that authorship be based on the following four criteria:

1. Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND
2. Drafting the work or revising it critically for important intellectual content; AND
3. Final approval of the version to be published; AND

4. Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

CRediT authorship contribution statement

Yong Wang: Methodology, Formal analysis, Data curation, Software, Writing – original draft, preparation. **Xinzhong Chen:** Supervision, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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